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ORIGINAL RESEARCH

# A small-target traffic sign detection algorithm based on partial conv and atrous spatial pyramid

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**Abstract**
Traffic sign detection is essential to an intelligent driving assistance system. The deep learning-based algorithm proposed in this paper aims to address the issue of low detection accuracy caused by the small size and high density of traffic signs in real-world traffic scenarios. First, to improve the feature extraction module of the backbone network and to increase the model's ability to capture contextual information, partial convolution (PConv) is introduced. Second, to prevent information loss during the downsampling process, a cross-stage atrous spatial pyramid (ASPPFCSPC) is constructed using atrous convolution. This method combines feature map information from various scales and expands the receptive field. Lastly, the small-target detection precision is improved by incorporating an additional small-target detection head, which uses high-resolution feature maps for shallow features. The detection head is decoupled to extract the location and class information of the predicted target separately, thereby enhancing the generalization ability of the proposed model. Experiments have demonstrated the superiority of the proposed algorithm, as testing on the TT100K dataset resulted in a mAP@0.5 of 91.2% and a mAP@0.5:0.95 of 71.8% using the proposed algorithm.

## 1 | INTRODUCTION

In autonomous driving scenarios, vehicles need to accurately and efficiently detect and understand input from the surrounding environment to ensure safety and compliance during driving. Traffic signs are a primary input source, as they convey a wide range of vital information to drivers and vehicles, including speed limits, road types, driving directions, and danger warnings. Therefore, accurate and efficient detection of traffic signs is essential to an automatic driving system.

The traditional traffic sign detection algorithm is divided into two steps. First, the region of interest is segmented in the image, and then the target is classified by combining a feature extraction method with a machine learning classifier. However, the manual feature characterization method used in traditional approaches has limitations and relies on expert knowledge. For large datasets, accurately distinguishing the subtle differences between fractional data is difficult, preventing use of such data in practical applications [1]. However, in recent years, the rapid development of deep learning technology has greatly advanced the progress of object detection. The mainstream traffic sign detection algorithm used currently, based on deep learning, can extract features from a large amount of data, yielding significant improvement in the level of accuracy achieved using traditional methods.

Due to the specificity of the traffic sign detection task, it is challenging to meet the practical requirements by using the existing object detection algorithms directly for this task. The existence of small targets significantly increases the difficulty of detection, as they occupy fewer pixels in the image and the feature information is relatively scarce [2]. This makes them susceptible to interference from background noise or other road elements, as well as making it difficult to maintain sufficient recognition in complex traffic scenes.

In order to address the issue of the instability of the detection effect of traffic sign detection algorithms and the difficulty of detecting occlusion and small targets, this paper proposes a small-target traffic sign detection algorithm based on improved YOLOv5. The main contributions of our work are summarized as follows:

1. Partial convolution (PConv) is implemented to improve the feature extraction module of the backbone network, which reduces model calculation parameters while ensuring detection accuracy. The convolution operation is used to simulate a multi-layer perceptron (MLP) structure, improving the performance of global feature extraction. Utilizing the cross stage partial network (CSPNet) structure for reference, the spatial pyramid pooling fast (SPPF) module is integrated to improve the spatial pyramid structure and atrous convolution is introduced to expand the receptive field, enabling better capture of long-distance dependence relationship in the image and strengthening the feature extraction capability of the network.
2. To improve the detection rate and accuracy for small targets, as well as enhance the feature extraction ability, a small-target detection head is added, fusing shallow and deep features. Furthermore, by decoupling the regression task from the classification task in the detection head, different network branches are used to learn the target location information and class information, so as to avoid mutual interference between different tasks, accelerate the convergence rate of the model, and improve the performance of different tasks.
3. The effectiveness of the proposed algorithm is verified on the TT100K dataset. Experiments have demonstrated that the proposed algorithm achieves competitive performance with mAP@0.5 of 91.2% and mAP@0.5:0.95 of 71.8%, which is 5.5% and 4.5% higher, respectively, than that of the benchmark algorithm. The proposed algorithm was tested on the real traffic scenarios dataset. The results show that the proposed algorithm can effectively reduce wrong detection and instances of failure to detect. Thus, the proposed algorithm can extract more effective feature information, and demonstrates good practicability and effectiveness in the traffic sign detection task.

The remainder of the paper is organized as follows: Section 2 describes the related works on traffic sign dataset, traffic sign detection, and small target detection. Section 3 provides a detailed description of the small target traffic sign detection network proposed in this paper. Section 4 analyses the experimental details and results. Finally, Section 5 describes the conclusions.

## 2 | RELATED WORKS

Researchers have studied traffic sign detection extensively and have proposed various methods to accomplish this task. In this section, we review early detection methods and recent state-of-the-art research in traffic sign detection and analyze their contributions and limitations. In Sections 2.1–2.3 we present traffic sign dataset, traffic sign detection and small target detection respectively.

### 2.1 | Traffic sign dataset

Datasets represent the fundamental resources that drive the training and optimization of deep learning models. In the context of the traffic sign detection task, the construction of a high-quality dataset is of particular importance, given the need to deal with various complex scenes and multiple sign types.

The German traffic sign detection benchmark (GTSDB) [3] is one of the most widely used benchmark in the field of traffic sign detection. However, the relatively limited data volume is not conducive to training deep learning models. The CSUST Chinese traffic sign detection benchmark (CCTSDB) [4] comprises nearly 20,000 images, which represents a considerable amount of data. However, the data categories are limited to three major categories of prohibition, warning, and indication, with no further subdivision. The images in the Tsinghua-Tencent 100K (TT100K) [5] comprises 30,000 instances of traffic signs, covering more than 200 different traffic sign categories. However, the distribution of data samples in this dataset is not uniform, and there are individual extreme samples.

Many researchers have carried out in-depth research into the problem of uneven sample data in traffic sign datasets. Luo et al. [6] combined two data acquisition methods—using street view images and synthetic images—to obtain a large number of labelled samples at a low cost. Igorevich et al. [7] proposed a neural network-based method for embedding new synthetic traffic signs in the road, and used the neural network with a spatial transformation module to predict the best position of additional signs. Jaghouar et al. [8] designed an iterative active learning algorithm to expand the training dataset according to the probability of each sign belonging to the rare class predicted by the network. However, relying solely on the algorithm to expand the data of rare classes causes a glaring problem. Some sample data generated by the algorithm is obviously contrary to basic common sense within a traffic context, which may lead to the acquisition of faulty information by the model.

## 2.2 | Traffic sign detection

The application of traffic sign detection technology has the potential to enhance the vehicle's perception of its surroundings, improve its understanding of road conditions, and thus optimize the overall driving experience. This field has attracted the attention of numerous scholars, who continue to explore and propose diverse detection algorithms with the objective of solving the challenges of traffic sign detection in an efficient manner. In particular, following the advent of deep learning, neural network-based algorithms for traffic sign detection have emerged as a research focus and the dominant approach in this field. The YOLO series algorithm [9–14] is the most widely used in this context. Yu et al. [15] proposed a fusion model based on YOLOv3 and VGG19 networks, which made full use of the relationship between multiple images to detect traffic signs in driving video sequences, but the model required a long training time. Wang et al. [16] proposed the attention-guided enhanced feature pyramid network (AF-FPN), which improves the detection performance of the YOLOv5 network, but the recognition effect is not satisfactory in situations such as image blur. Zhang et al. [17] enhanced YOLOv8 by introducing an attention mechanism and the receptive field block (RFB) module to improve channel and spatial features. The aforementioned methodologies markedly augment the number of model parameters, rendering them challenging to fulfil the real-time necessities of the traffic sign detection task.

At the same time, the researchers also used deep learning technology to build an adaptive network with a stronger generalization ability to cope with traffic sign detection tasks under different scenarios and conditions. Hu et al. [18] proposed a traffic sign detection and recognition paradigm based on a deep learning algorithm in [19], introduced deformable convolution, improved model feature extraction ability, resulting in better retained spatial information. Yan et al. [20] proposed an adaptive enhancement algorithm for traffic sign detection under complex lighting conditions, which can effectively improve the image quality of subsequent detection. Li et al. [21] proposed an attention-guided context feature pyramid network (ACFPN), which is used to balance input feature resolution and the receptive field. To explore the relationship between traffic sign detection performance and driving time in a hazy environment, Hu et al. [22] proposed a UV model that is associated with driving sight distance, haze level, and traffic sign detection performance.

## 2.3 | Small target detection

Small target detection refers to the process of identifying and localizing objects in an image or video frame that are of a microscopic size and occupy a limited number of pixels. Such targets are frequently challenging to distinguish effectively from complex backgrounds due to the scarcity of visual information and the interference of scale variations. Given the small size of traffic signs and their limited proportion of the captured image, coupled with the complexity and variability of the actual traffic scene and potential occlusion, the traffic sign detection has become particularly challenging.

For the problem of small target detection, researchers have conducted in-depth research and exploration. Li et al. [23] introduced the self-attention mechanism and convolutional mixing module ACMix [24] to improve the network structure, and introduced normalized Wasserstein distance (NWD) [25] to measure and enhance the sensitivity to small-target regression bias. Wang et al. [26] took advantage of FasterNet [27] to propose an efficient feature processing module named focal FasterNet block (FFNB). This method integrates both shallow and deep features and significantly reduces the missed detection rate of small targets. Qu et al. [28] adopted YOLOv5 as the benchmark model, integrated the coordinate attention (CA) mechanism [29] to filter redundant feature information, and added a small-target detection layer to improve the overall accuracy of the model. Li et al. [30] proposed the occlusion-aware module, which exploits the sensitivity of average peak-to correlation energy to occlusion in order to reduce the impact of occluded scenes on small target detection. Wang et al. [31] presented a query-based model for the purpose of filtering out the least relevant key-value pairs at a coarse region level. This is done in order to reduce the size of the routing region, thereby improving the detection performance of remote small targets. Zhang et al. [32] proposed a feature fusion module (FFM) that adapts the original BiFPN to fit three detection heads and enhances the CRC reweighting strategy, thereby improving the weak feature representation of small objects. However, the above methods are all targeted at specific datasets and lack generalization verification for real-world scenarios.

# 3 | METHOD

## 3.1 | The overall framework of proposed algorithm

The overall network structure of the proposed algorithm in this paper is shown in Figure 1, divided into three parts: backbone, neck, and head. The network's foundation module, termed CBS, includes a standard convolutional layer, a batch normalization layer, and a sigmoid linear unit (SiLU) activation function. The backbone network essentially combines the CBS module, a PC3 module with PConv, and a cross-stage atrous spatial pyramid (ASPPFCSPC) for feature extraction to strengthen the model's understanding of semantic information and effectively integrate spatial information. The neck network combines the feature pyramid network (FPN) and path aggregation network (PANet) to obtain multi-scale features. The head network uses a four-scale decoupling detection head to detect the extracted features and enhance the sensitivity of small-target details. The decoupled structure splits the classification and regression tasks to prevent interaction between different tasks.

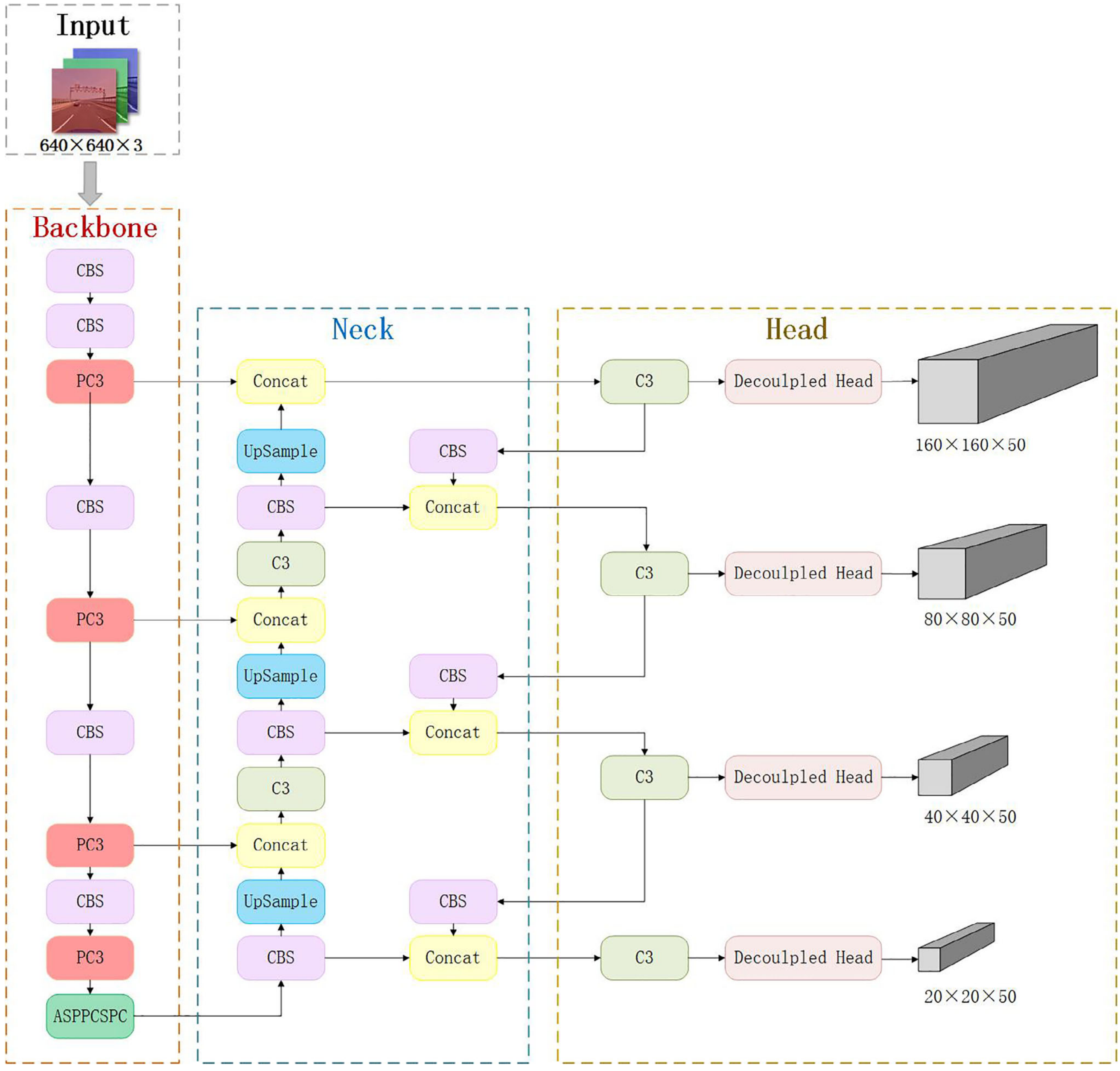



**FIGURE 1** Overall structure of the proposed network.

## 3.2 | Backbone

### 3.2.1 | PC3 module

As illustrated in Figure 2, the C3 module of YOLOv5 employs a two-branch structure. Each branch is processed by a CBS module, reducing the number of channels by half. Subsequently, one branch executes the Bottleneck operation, and the two branches are concatenated to maintain consistency in input and output dimensions. This design enables the model to acquire a greater number of features, although this does entail an increase in the amount of computation required by the model. To address this issue, this paper proposes PConv, a method that can more effectively extract spatial features by concurrently reducing superfluous computation and memory access.

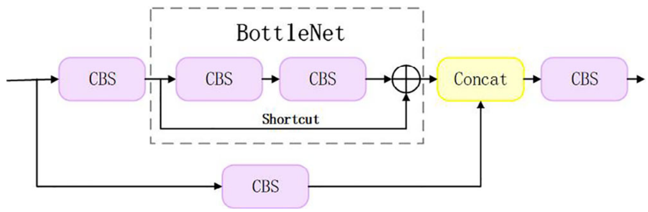



**FIGURE 2** Structure of C3 module.

PConv uses the feature similarity between different channels to learn features only for some channels, while the other

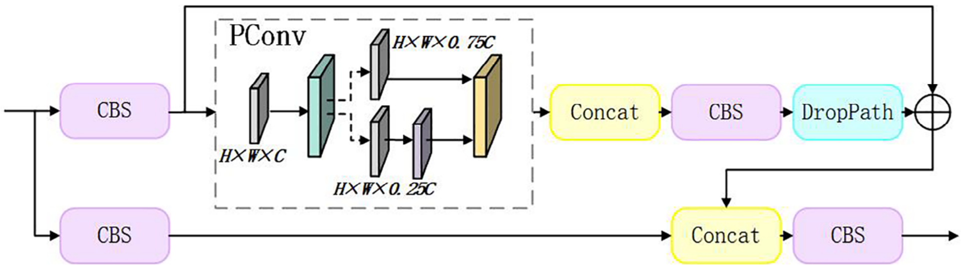



**FIGURE 3** Structure of PC3 module.

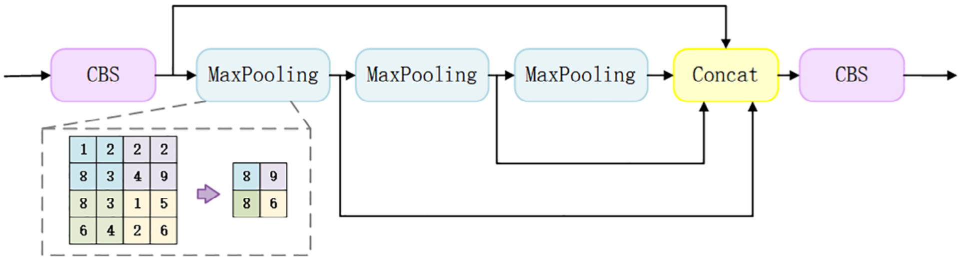



**FIGURE 4** Structure of SPPF module.

channels remain unchanged. In this paper, the algorithm introduces PConv to improve the original standard convolution layer and constructs the PC3 module using convolution layers to simulate the MLP structure and combine local features into global features. Compared with the direct use of the full connection layer, the number of parameters is effectively reduced, and the computational efficiency of the convolution operation is retained.

The improved PC3 module structure is shown in Figure 3. The PC3 module is the main module for learning residual features. As shown in Figure 3, the input feature map is processed through two different branches, in which the main path is processed by multiple MLPBlock stacking, and the residual connection is processed by only a 1 × 1 standard convolution module; then the output of the two branches is cascaded together. The MLPBlock module also adopts a residual structure, and the main path processing module includes a PConv module, a CBS module, and an MLP module simulated by standard convolution layer, which is then regularized by DropPath. The addition of a residual structure can effectively solve the problem of gradient vanishing and gradient exploding, and also retain the original input feature image. PConv uses only part of the channels of the feature map to extract spatial features, and the remaining channels do not carry out any operations, which ensures efficiency and detection accuracy while reducing calculation parameters.

### 3.2.2 | Cross-stage atrous spatial pyramid

In the original model of YOLOv5, the spatial pyramid pooling fast (SPPF) structure is adopted to complete the task of aggregating features on multiple scales. As illustrated in Figure 4, in the SPPF module, the input feature map is subjected to a CBS module for feature compression. This is followed by the implementation of three max pooling operations in a serial manner, after which feature map fusion is conducted on the results of the three pooling operations and the result of the feature compression. Finally, the CBS module is employed to complete the dimensionality enhancement.

The SPPF structure uses the max pooling layer to downsample the feature image, selectively retain the information, enlarge the receptive field, and fuse the feature map information of different scales. However, when the pooling layer expands the receptive field, it will reduce the spatial resolution, resulting in the loss of the feature map details, and the pixel-level deviation of the feature map will cause detection errors and reduce the accuracy of network detection. In order to solve this problem, this paper introduces the atrous convolution to construct the cross-stage atrous spatial pyramid termed the ASPPFCSPC.

Figure 5 shows the structure of the ASPPFCSPC module, which adopts the CSP [33] structure. First, the features are divided into two parts: one part directly adjusts the number of channels through the 1 × 1 convolution module, and the other part is processed by the atrous spatial pyramid. Then the two parts are combined to promote information flow and gradient propagation. It also helps to alleviate the problem of gradient vanishing and gradient exploding.

SPPF processes the input feature map serially through multiple maximum pooling layers, and each pooling layer performs pooling operations on the output of the previous pooling layer to achieve feature fusion at different scales. The improved module uses atrous convolution instead of the pooling layer for processing. Compared with the pooling operation, atrous convolution can enlarge the receptive field without losing the resolution, and keep the relative spatial position of pixels unchanged, reducing the loss of image information.

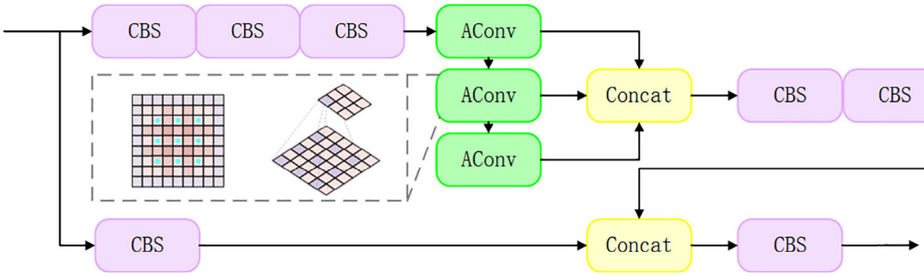



**FIGURE 5** Structure of ASPPFCSPC module.

## 3.3 | Head

### 3.3.1 | Small-target detection head

Traffic signs usually occupy a relatively small visual space in real-world traffic scenarios. The detection heads of three different scales designed in the existing network structure carry out 8×, 16×, and 32× downsampling for the original image by default, and lose more original image information after multiple down-sampling, which works against the small-target detection task. To solve this problem, this paper proposes adding a small-target detection head to enhance the sensitivity of the model to small targets. The additional detection head is generated by the shallow high-resolution feature map and is more sensitive to small targets, which can effectively improve the detection efforts of the model and decrease small-targets detection failures.

### 3.3.2 | Decoupled detection head

The main task of the head network is to complete the target detection according to the extracted features and output the object class, the object location, and the detection confidence. In the traditional target detection algorithm, the output feature map is usually directly sent into several fully connected layers or convolutional layers to generate the target position and class output. However, due to the different optimization objectives and features of different tasks [34] shared features are not applicable to all tasks, resulting in the model learning some shared representations with poor generalization effect, which is prone to overfitting. Therefore, the algorithm proposed in this paper decouples the classification and regression tasks, extracts the target location and class information separately, learns through different network branches, and finally merges, so that the network can independently learn and adjust the model parameters for different tasks, and thus improve the performance of the model.

The algorithm proposed in this paper uses decoupled detection heads of different sizes to detect targets for feature graphs of different scales. Figure 6 shows the structure of a single decoupled detection head. Each decoupled detection head contains a CBS module to reduce the dimension of the input feature image to reduce the amount of computation, and then the processing structure is divided into two branches according to different tasks. The classification branch mainly focuses on the class information of the target, while the regression branch gathers the target's location information. The regression branch further splits to capture the position regression of the detection box and the target confidence prediction. Finally, the three branch outputs are concatenated according to the channel dimension to obtain the result.

## 4 | EXPERIMENTS

In this section, the experimental environment and adopted dataset are presented along with common object detection model performance evaluation indicators used to evaluate the proposed model. Comparative experimental results are analysed based on the TT100K dataset to verify the effectiveness of the proposed algorithm, and ablation experimental results are displayed to demonstrate the importance of each module in the proposed architecture. Finally, the effectiveness of the algorithm is verified using the publicly available dataset and real-world traffic scenarios.

## 4.1 | Environment and dataset

### 4.1.1 | Implementation details

The hardware used for the experiments included the 12th Generation Intel Core i7-12700 and NVIDIA GeForce RTX 3060Ti; Pytorch 2.0.1 was used for the deep learning framework. The specific settings of network training hyperparameters are as follows: the initialization of the training model was carried out using the official YOLOv5s.pt weight file, the batch size was set to 4, the number of iterations was set to 600, and the input image size was set to 640 × 640.

### 4.1.2 | Dataset

For this paper, the public traffic sign dataset TT100K was adopted as the experimental data, as it contains more than 100 different classes of traffic signs. However, given the unbalanced data distribution and the low occurrence rate of some traffic signs in real-world traffic scenarios, 45 types of traffic signs with sufficient sample data (more than 100 samples) were selected as research objects. Table 1 shows the detailed class labels used

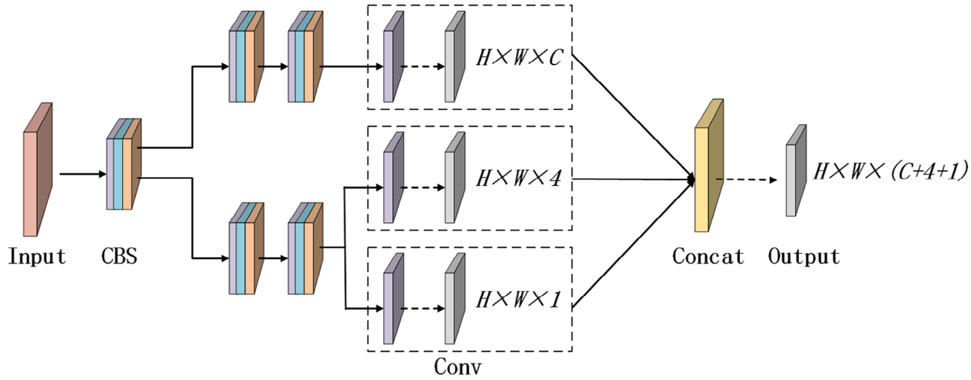
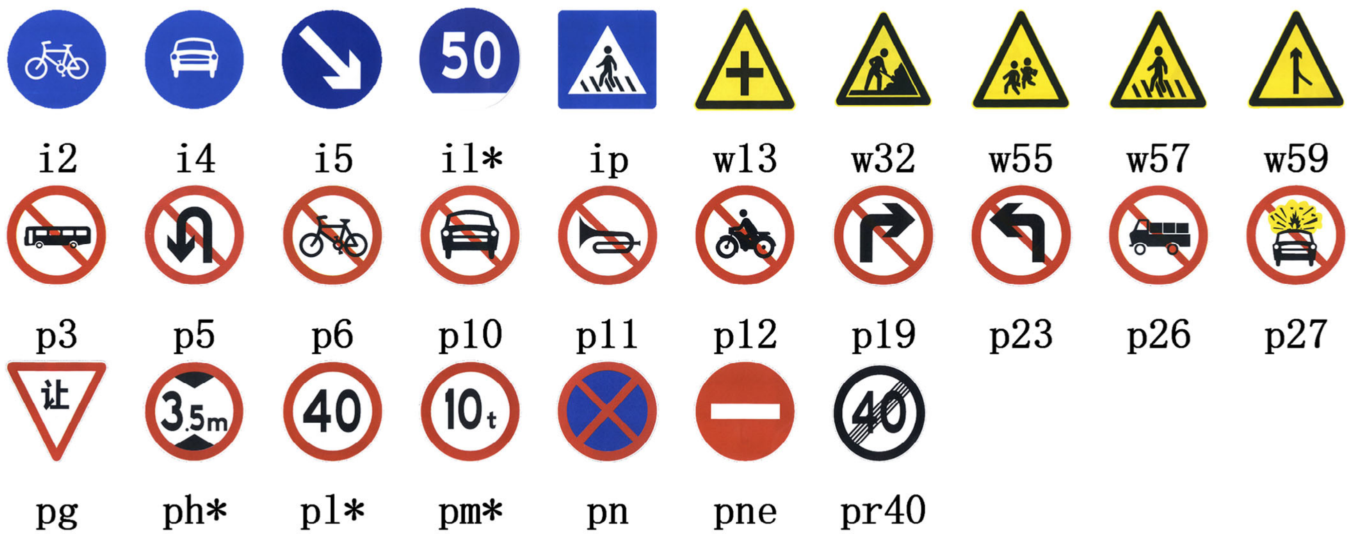



**FIGURE 6** Decoupled detection head structure.




Note: * indicates the number shown in the symbol

**FIGURE 7** Mapping between labels and legends.

**TABLE 1** Detailed class labels.

| Class | Label Name |
|---|---|
| Indication | i2, i4, i5, il100, il60, il80, io, ip |
| Warning | w13, w32, w55, w57, w59, wo |
| Prohibit | p3, p5, p6, p10, p11, p12, p19, p23, p26, p27, pg, pn, pne, po, pl100, pl120, pl20, pl30, pl40, pl5, pl50, pl60, pl70, pl80, ph4, ph3.5, ph5, pm20, pm30, pm55, pr40 |

in the experiment, and the corresponding specific labels and legends are shown in Figure 7. It should be noted that the categories io, wo, and po pertain to signs that resemble traffic signs but do not convey the same meaning. Consequently, they are not depicted in the Figure 7.

## 4.2 | Evaluation criteria

To prove the advantages of the proposed algorithm in traffic sign detection, the performance of the model was evaluated using common object detection evaluation indexes such as precision (P), recall (R), F1-score, average precision (AP), mean average precision (mAP), and frames per second (FPS).

The precision index measures the accuracy of the model when the prediction is positive, and recall measures the sensitivity of the model to positive samples. The calculation formula for these two indexes is shown in Equations (1) and (2).

$$\text{Precision} = \text{TP}/\left(\text{TP} + \text{FP}\right) \tag{1}$$

$$\text{Recall} = \text{TP}/\left(\text{TP} + \text{FN}\right), \tag{2}$$

where TP (true positives) indicates the number of positive samples predicted as positive cases, TF (true negatives) indicates the number of negative samples predicted as negative cases, and FP (false positives) indicates the number of negative samples wrongly predicted as positive cases. FN (false negatives) indicates the number of positive samples that are incorrectly predicted to be negative.

In order to achieve an optimal balance between the precision and recall of model, the F1-score is introduced as an evaluation index. This score is calculated by combining the precision and recall rates into a single score, which takes into account the effects of false positives and false negatives. The calculation of the F1-score is based on the harmonic mean of the precision and recall. The formula for calculating the F1-score is shown in Equation (3).

$$\text{F1} - \text{score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) \tag{3}$$

AP represents the calculation of the accuracy of the detection results when they match the true label for a specific category, with these values being averaged. The calculation formula is shown in Equation (4).

$$AP = \int_0^1 P(R)\,\mathrm{d}R\,, \tag{4}$$

The mAP value takes into account the performance of the model in all classes, and is a comprehensive evaluation index for tasks such as object detection. The calculation formula is shown in Equation (5).

$$\text{mAP} = \frac{1}{n}\left(\sum_{i=1}^{n}\int_0^1 P(R)\,\mathrm{d}R\right), \tag{5}$$

where $n$ represents the number of classes in the task. In the evaluation of object detection algorithms, the mAP@0.5 and mAP@0.5:0.95 metrics are frequently employed to represent the mAP at varying intersection over union (IoU) thresholds. The IoU is the overlap metric between the predicted and actual bounding boxes. The mAP (mean average precision) at an IoU threshold of 0.5 is denoted by mAP@0.5. The mAP@0.5:0.95 denotes the average mAP computed at multiple IoU thresholds ranging from 0.5 to 0.95 with a step size of 0.05.

Accuracy is a very basic and important evaluation metric in classification tasks. The calculation formula is shown in Equation (6).

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \tag{6}$$

In order to demonstrate the enhanced performance of the proposed method for the detection of small target traffic signs, the COCO evaluation metrics $AP_s$, $AP_m$, $AP_l$, and $AR_s$, $AR_m$, $AR_l$ are introduced in this paper for the comparison of the detection accuracy and recall of targets of varying sizes. It is important to note that the criteria for dividing the various different sized targets are as follows:

1. Small objects: area $<32^2$
2. Medium objects: $32^2<$ area $<96^2$
3. Large objects: area $>96^2$

FPS is employed to assess the speed of model detection, specifically the number of images that can be processed in a given time period or the time required to process an image. A shorter processing time indicates a faster speed, which is a crucial factor in achieving real-time detection.

**TABLE 2** Comparative experimental results with other generic detectors.

| Model | P (%) | R (%) | F1-score | mAP @0.5 (%) | mAP@0.5: 0.95 (%) | FPS |
|---|---|---|---|---|---|---|
| YOLOv5s | 86.0 | 80.1 | 82.95 | 85.7 | 67.3 | 163.93 |
| YOLOv5m | 88.3 | 84.4 | 86.31 | 88.5 | 69.6 | 104.17 |
| YOLOv6s | 83.7 | 73.6 | 78.33 | 82.3 | 64 | 143.88 |
| YOLOv7 | 86.8 | 83.7 | 85.22 | 90.7 | 71.6 | 51.02 |
| YOLOv8n | 79.7 | 70.7 | 74.93 | 78.7 | 61.3 | **200.00** |
| YOLOv8s | **88.5** | 76.4 | 82.01 | 86.9 | 69.2 | 136.99 |
| Proposed | 87.9 | **86.2** | **87.04** | **91.2** | **71.8** | 48.78 |

## 4.3 | Comparative experiments analysis

### 4.3.1 | Comparison with other generic detectors

To verify the advancement of the improved YOLOv5 algorithm, comparative experiments were conducted based on the TT100K dataset, and the mainstream single-stage object detection models such as YOLOv5, YOLOv6, YOLOv7, and YOLOv8 were compared with the algorithm proposed in this paper. The experimental results are shown in Table 2.

As Table 2 demonstrates, the algorithm proposed in this paper achieves the best mAP results, with mAP@0.5 reaching 91.2% and mAP@0.5:0.95 reaching 71.8%. Compared with other generic detectors (YOLOv5s, YOLOv5m, YOLOv6s, YOLOv7, YOLOv8n, and YOLOv8s), the average accuracy of the algorithm proposed in this paper is as follows: mAP@0.5 is higher by 5.5%, 2.7%, 8.9%, 0.5%, 12.5%, and 4.3% respectively, and mAP@0.5:0.95 is higher by 4.5%, 2.2%, 7.8%, 0.2%, 10.5%, and 2.6% respectively. The comparison results show that the proposed algorithm has significant advantages in performance. Although there is a reduction in the speed of detection, the FPS was still able to meet the real-time detection requirements.

During the training process, the average accuracy of each algorithm model mAP@0.5 varies with the number of training iterations, as shown in Figure 8. In the early training period, the mAP value of each model increases rapidly; when the training times reach about 300, the growth rate of the curve decreases; when the training times reach about 500, the curve gradually becomes stable. As from Figure 8 illustrates, the proposed algorithm achieves good performance on the TT100K dataset, and the mAP@0.5 stabilizes at 91.2%, which is obviously superior to the results of the other algorithms.

### 4.3.2 | Comparison with other traffic sign detectors

The proposed method was evaluated in comparison with other traffic sign detection algorithms on the TT100K dataset. As

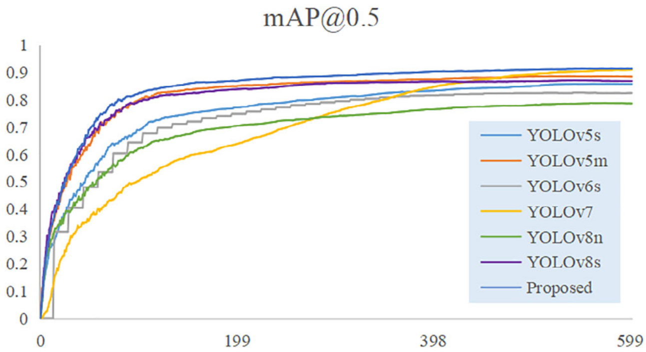



**FIGURE 8** The average accuracy of each model with the number of iterations.

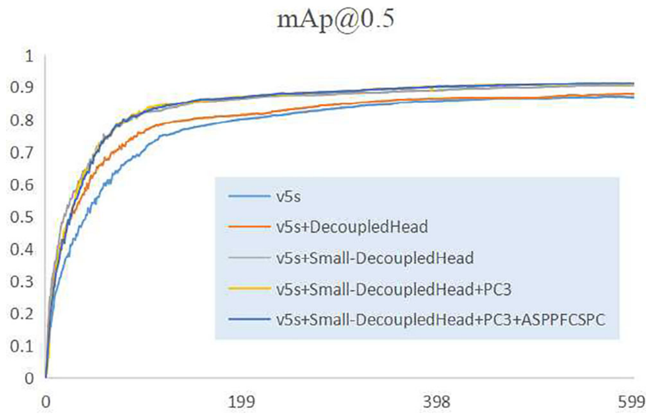



**FIGURE 9** The average accuracy of each experiment with the number of iterations.

**TABLE 3** Comparative experimental results with other traffic sign detectors.

| Model | F1-Score | mAP@0.5 (%) |
|---|---|---|
| ETSR-YOLO [35] | 86.17 | 88.3 |
| YOLO-SG [36] | 71.00 | 75.8 |
| IYOLOv4 [37] | \ | 87.3 |
| Zhang et al. [38] | \ | 70.16 |
| TSDet [39] | \ | 75.3 |
| Li et al. [40] | \ | 90.1 |
| TSD-YOLO [41] | 85.75 | 90.6 |
| Proposed | **87.04** | **91.2** |

the majority of paper codes are not open source, the data presented in their papers serve as a reference point. The experimental dataset for each model is based on TT100K. The results are presented in Table 3. The proposed method demonstrated the highest performance, achieving F1-score of 87.04 and mAP@0.5 of 91.2%.

## 4.4 | Ablation experiments analysis

To achieve accurate and efficient detection of small-target traffic signs, ablation experiments were conducted based on the TT100K dataset to analyze the importance of each module. Using YOLOv5s as the basic model, the decoupled detection head, small-target detection head, PC3, and ASPPFCSPC were added to the model step by step to form multiple improved models. The experimental results are shown in Table 4.

As Table 4 demonstrates, adding the decoupled detection head, small-target detection head, PC3, and ASPPFCSPC to the basic model of YOLOv5 can effectively improve the model's mAP. This result is achieved because the decoupled detection head splits the classification and regression tasks, and each task uses an independent network structure, which helps the model to better learn task-specific features and improve task performance. The small-target detection head can better learn the small sample features in the dataset and enhance the sensitivity of the model to small targets. PConv can extract spatial information more effectively through local convolution operation, thus improving the performance of neural networks. ASPPFCSPC can capture different scale information of the input feature map to improve the detection performance of objects of different sizes.

As shown in Table 4, Method A is the basic model of YOLOv5s. Method B added the decoupled detection head to the original model to separate classification and regression tasks, improving mAP@0.5 by 2.2% and mAP@0.5:0.95 by 0.7%, indicating that these tasks require distinct features. Method C added a small-target detection head to enhance the sensitivity of the model to small targets, and mAP@0.5 improved by 4.7% and mAP@0.5:0.95 improved by 3.3% compared with the original model. In method D, the PC3 module replaced the C3 module in the backbone network, with mAP@0.5 improved by 5.4% and mAP@0.5:0.95 improved by 3.7%, indicating that PC3 helps to better retain the structure and information of images. Method E used the ASPPFCSPC module to replace the original SPPF pyramid module, which enhanced the model's adaptability to targets of different sizes; mAP@0.5 reached 91.2%, which was 5.5% higher than the original model, and mAP@0.5:0.95 was 4.5% higher than that of other improved models. The combination of the AP and AR evaluation metrics for each size demonstrates that Method E performs robustly when dealing with smaller sized objects (56.0% for $AP_s$), and also exhibits high accuracy on medium and large sized objects (78.7% for $AP_m$ and 83.8% for $AP_l$). Furthermore, it is evident that Method E significantly outperforms the other methods on the AR metrics for each scale. During the training process, the average accuracy of each algorithm model mAP@0.5 varies with the number of training iterations, as shown in Figure 9.

## 4.5 | Algorithm effectiveness analysis

### 4.5.1 | Validity verification of public dataset

To evaluate the performance of the algorithm proposed in this paper, YOLOv5s, YOLOv5m, and the proposed algorithm were used to conduct subjective visual comparison under

**TABLE 4** Results of ablation experiments.

| Method | ① | ② | ③ | ④ | Accuracy | mAP@0.5 | mAP@0.5:0.95 | APs | APm | APl | ARs | ARm | ARl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | | | | | 74.7 | 85.7 | 67.3 | 50.7 | 74.8 | 79.3 | 63 | 82.2 | 88.2 |
| B | √ | | | | 75.5 | 87.9 | 68 | 52.7 | 75.2 | 79.5 | 62.3 | 81.9 | 89.4 |
| C | √ | √ | | | 76.8 | 90.4 | 70.6 | 56.2 | 76.4 | **86.3** | 68.1 | 83.7 | 88.9 |
| D | √ | √ | √ | | 77.0 | 91.1 | 71 | **56.3** | 77.3 | 84.1 | 67.9 | 83.7 | 90.2 |
| E | √ | √ | √ | √ | **78.2** | **91.2** | **71.8** | 56.0 | **78.7** | 83.8 | **68.2** | **84** | **90.5** |

*Note*: (1) The "√" symbol indicates the addition of this component to the YOLOv5 model schema.
(2) The indexes in the table correspond to the components as follows: ① decoupled head, ② small-target head, ③ PC3, ④ ASPPFCSPC.
(3) All data presented in the table are expressed as a percentage.

different traffic scenarios, as shown in Figure 10. Figure 10 shows the test results of the three different algorithms on TT100K test dataset. The first column of data is the original image and labels, and the following three columns of data are the results of detection using YOLOv5s, YOLOv5m, and the proposed algorithm.

As shown in Figure 10, in the two groups of Figures 10a and 10d, YOLOv5s could not detect remote dense targets pl50, p11, and pn, while YOLOv5m could not detect remote p11 and pn, and the detection effect of remote pl50 targets was unstable. In Figure 10b, YOLOv5s and YOLOv5m fail to detect class il80, and in Figure 10c, YOLOv5s fails to detect class ip. However, the proposed algorithm can accurately detect all targets with almost no misses and false detections, and the detection confidence is higher. The analysis results show that the algorithm proposed in this paper can successfully detect small-target traffic signs, reduce the false detection rate of the original model, and produce a high rate of accuracy in detection.

### 4.5.2 | Validity verification of real scenarios

To verify the detection capabilities of the proposed algorithm in real-world traffic scenarios, a vehicle-mounted camera was used to collect a road scenario test dataset using images of Hedi Road in Xixian New Area, Xi'an, Shaanxi Province. YOLOv5s, YOLOv5m, and the proposed algorithm were used to conduct the performance test on the self-built dataset. The results are shown in Figure 11. Figure 11 shows the test results of three different algorithms on the real-world traffic scenarios. The first column of data is the original image and labels, and the following three columns of data are the results of data detection by YOLOv5s, YOLOv5m, and the proposed algorithm.

As shown in Figure 11, in the two groups of Figures 11a and 11d, YOLOv5s showed false detection for class pl60, while YOLOv5m showed unstable and false detection performance for this class. In Figure 11b, YOLOv5s and YOLOv5m missed detection for class wo. In Figure 11c, YOLOv5s failed to detect class pl40, and YOLOv5m failed to detect class pl40. In the two sets of Figures 11e and 11f, YOLOv5s failed to detect classes ip and pn. Although YOLOv5m detected the target, the confidence was low. However, the algorithm proposed in this paper demonstrated that it can accurately and quickly detect the traffic signs in different traffic scenarios, and has a strong generalization ability.

## 5 | CONCLUSIONS

Traffic sign detection is a challenging task for intelligence driving systems due to the small size and high density of traffic signs in real-world traffic scenarios, as well as the complex and changeable environment that is subject to extreme weather, variable illumination, and other dynamic conditions. To address these challenges, this paper proposes a small-target traffic sign detection algorithm based on improved YOLOv5. Partial convolution is introduced first, and the feature extraction structure of the network is improved by incorporating a multi-layer perceptron, which strengthens the global feature extraction capability and reduces the computation, improving the real-time performance of the model. Using a spatial pyramid and CSPNet network structure as a reference, the ASPPFCSPC module is constructed to integrate feature information of different scales and reduce information loss during feature extraction. To further improve the model's detection of small-target traffic signs, a detection head is added to identify objects of different scales. Finally, the decoupled detection head is introduced to extract the target position information and class information, further improving the model's performance.

The experimental results show that, compared with other mainstream object detection algorithms, the proposed algorithm has excellent performance in various evaluation indicators and effectively improves the detection accuracy of traffic signs in complex road scenarios. The test results on the TT100K dataset and self-built dataset fully reflect its superior detection performance. Nevertheless, the algorithm proposed in this paper still exhibits certain deficiencies.

1. The proposed solution does not address the fundamental issue of the uneven distribution of traffic sign data.
2. In order to enhance the precision of detection, the majority of algorithms employ an incremental expansion of the number of convolutional neural network layers, which


**FIGURE 10** Original image and detection results of three algorithms on the TT100K dataset.

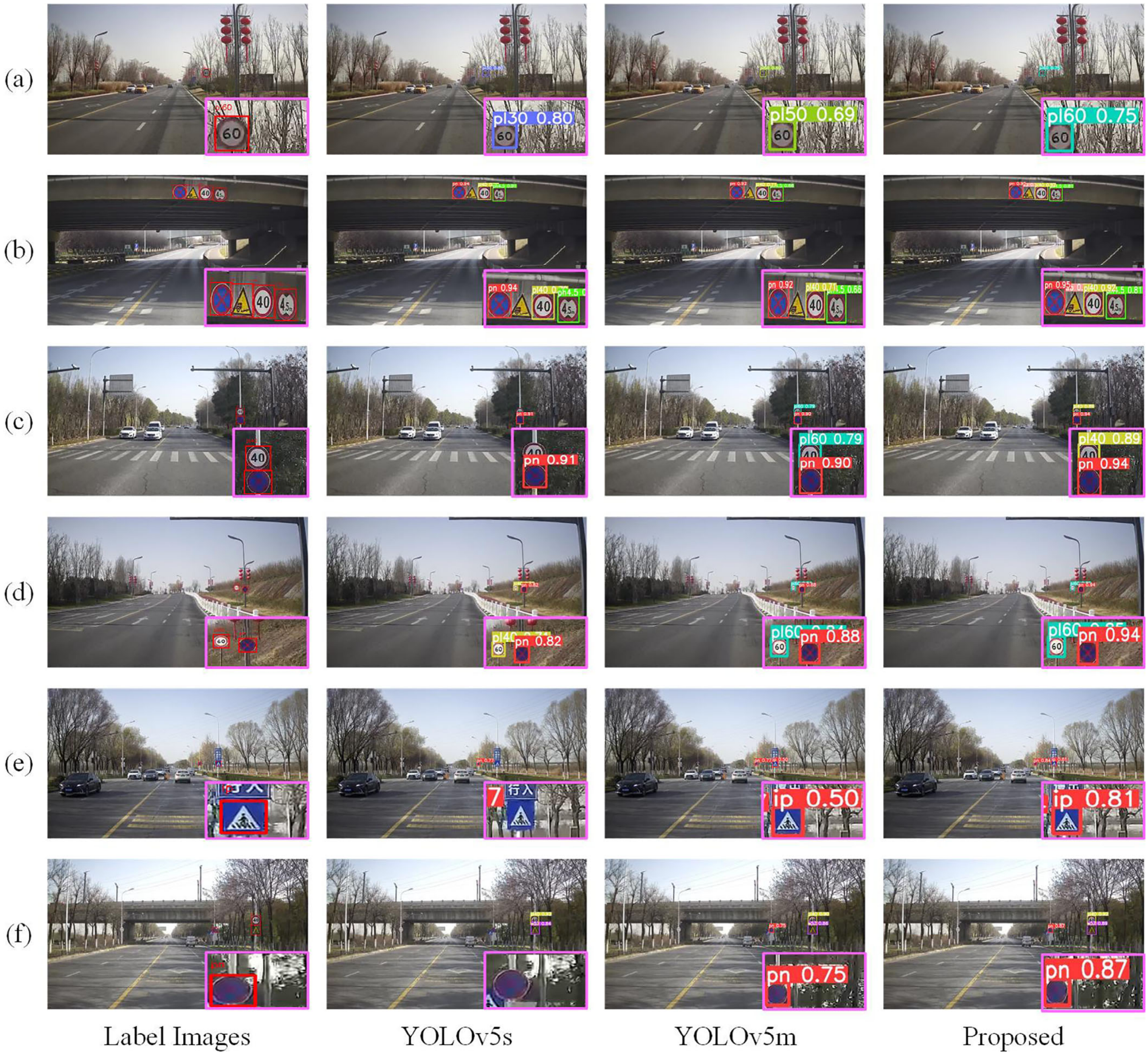



**FIGURE 11** Original image and detection results of the three algorithms on the self-built dataset.

consequently leads to an augmented computational burden and a diminished real-time functionality [42].

On the basis of this work, future research can focus on the development of models that are both lightweight and accurate. In addition, zero-shot [43] or few-shot [44] detection methods could be considered in order to accomplish this task.

## AUTHOR CONTRIBUTIONS

**Yuqi Li**: Methodology; software; writing—original draft; writing—review & editing. **Zijian Wang**: Investigation; funding acquisition; data curation. **Han Zhang**: Investigation; visualization. **Xinpeng Yao**: Resources; formal analysis. **Zhou Zhou**: Validation; funding acquisition. **Xin Cheng**: Conceptualization; funding acquisition; supervision.

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## CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

## DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in TT00K at https://cg.cs.tsinghua.edu.cn/traffic-sign/.

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